

Exception Flow Documentation

Customizable Subscription Box Scheduler — Problem Statement #35

Use Case: Customize Monthly Box

ID: UC-002

Primary Actor: Subscriber

Secondary Actor(s): System (Recommendation Engine, Fulfillment Service, Payment Gateway)

Related Requirements: FR-001, FR-002, NFR-001, NFR-002

Purpose: This document captures the failure conditions that can interrupt the Main Success Scenario or Alternative Flows of UC-002 — conditions caused by unmet preconditions, invalid data, external system failures, or infrastructure faults — along with the system's response and recovery behavior for each.

See the accompanying file [Exception_Flow_Diagram_UC-002.png](#) for the visual diagram mapping each exception to the step where it can be triggered.

Exception Flows

EF-1 — Recommendation Engine Unavailable	
Triggered At	Step 3 — «include» Apply Preference-Based Recommendations
Trigger Condition	The Recommendation Engine service does not respond within the expected timeout, or returns an error.
System Response	System catches the failure and falls back to displaying the Subscriber's last saved box contents plus the full product catalog (without preference-match scoring), and shows a banner: "Personalized picks are temporarily unavailable — browse the full catalog instead."
Subscriber Recovery	Subscriber may still manually browse and swap items using the catalog; the flow rejoins the Main Success Scenario at Step 5.
Related Requirement	FR-002 (degrades gracefully rather than blocking customization)

EF-2 — Replacement Item Out of Stock	
Triggered At	Step 6 — System validates each replacement is in stock
Trigger Condition	Subscriber selects a replacement item whose inventory reaches zero between page load and selection (race condition) or was never in stock.
System Response	System rejects the specific swap, greys out the item in the UI, and displays: "This item just sold out — please choose another." All other valid selections in the same session are preserved.
Subscriber Recovery	Subscriber selects a different replacement; flow resumes at Step 6 for the corrected item.
Related Requirement	FR-001 (acceptance criteria: customization must reflect real-time stock)

EF-3 — Item Incompatible with Subscription Tier	
Triggered At	Step 6 — System validates each replacement is compatible with the subscriber's subscription tier
Trigger Condition	Subscriber selects an item that is reserved for a higher subscription tier than their current plan.
System Response	System rejects the swap and displays: "This item is available on the Premium tier. Upgrade to add it to your box." with a link to plan upgrade options.

Subscriber Recovery	Subscriber either selects a tier-eligible item and resumes at Step 6, or exits the flow to upgrade their subscription (use case ends, no box changes persisted).
Related Requirement	FR-001

EF-4 — Session or Authentication Expires Mid-Customization	
Triggered At	Steps 5–8 — while the Subscriber is actively reviewing, selecting, or confirming changes
Trigger Condition	The Subscriber's authenticated session times out or is invalidated (e.g., login elsewhere) before "Save My Box" is clicked.
System Response	System holds the in-progress (unsaved) selections client-side for up to 5 minutes and redirects to the login page with: "Your session expired. Log in to continue where you left off."
Subscriber Recovery	On successful re-authentication within the hold window, the System restores the in-progress selections and resumes at Step 7. If the window lapses, the Subscriber restarts from Step 1 with the box reverted to its last saved state.
Related Requirement	NFR-001 (authenticated, role-appropriate access only)

EF-5 — Price-Difference Authorization Fails	
Triggered At	Steps 7–8 — preview shows a price difference and Subscriber confirms "Save My Box"
Trigger Condition	A selected replacement costs more than the item it replaces, and the Subscriber's payment method on file is declined or expired when the system attempts pre-authorization.
System Response	System reverts the affected swap(s) to the original items, keeps any same-price or lower-price swaps intact, and displays: "We couldn't authorize the price difference for [item]. Please update your payment method or choose a different item."
Subscriber Recovery	Subscriber updates the payment method or selects a compatible-price item, then resumes at Step 7 to re-confirm.
Related Requirement	NFR-001 (secure transmission of payment-related data)

EF-6 — System or Network Failure During Save	
Triggered At	Steps 9–10 — persisting selections and refreshing the fulfillment manifest
Trigger Condition	A database, network, or service failure occurs after the Subscriber confirms changes but before the transaction fully commits.
System Response	System rolls back the transaction to the last consistent state (no partial writes), releases any inventory holds acquired during this attempt, and displays: "Something went wrong saving your box. Your previous box is unaffected — please try again."
Subscriber Recovery	Subscriber retries "Save My Box"; if the failure persists, Subscriber is advised to retry later, and the previously saved box ships as a safe fallback.
Related Requirement	NFR-002 (99.9% uptime; stable behavior under failure)

EF-7 — Concurrent Modification Conflict	
Triggered At	Step 9 — System persists updated selections

Trigger Condition	The same Subscriber (e.g., logged in on two devices) submits conflicting changes to the same upcoming box in overlapping sessions.
System Response	System accepts the first successful save, then rejects the second with: "Your box was updated in another session. Refresh to see the latest version before making further changes."
Subscriber Recovery	Subscriber reloads the "My Upcoming Box" page to see the currently saved state and, if desired, restarts customization from Step 1.
Related Requirement	FR-001 (data integrity of the customization workflow)

Note: *In every exception flow, no partial or inconsistent box state is ever persisted or shipped — the Subscriber's last successfully saved box remains the fallback until a new save succeeds.*